# Machine learning in real companies

Real use cases, concepts and trade-offs

InnoIT · 19 October 2026



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# Learning path

**01.** Machine learning foundations

**02.** Deep learning and image generation

**03.** GenAI systems and agentic workflows

12 requested topics. 15 cases, a solution diagram for each, and an integrated BBVA Blue closing example.

# A guide to the examples

## Tasks and implementations

Classification, clustering and forecasting describe tasks. Several algorithm families can solve each one.

RAG and agent orchestration describe systems that combine models with other software.

## How to read each case

Each case covers the concept, real workflow, advantages, limitations and a one-sentence takeaway.

Sources describe the documented deployment at the time of publication. Trade-offs are teaching analysis.

# 01

# Machine learning foundations

Relationships, labels, groups and numerical predictions



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# Association rules

## Walmart (United States)

Production recommendation system

### Concept

Find items that appear together more often than chance would suggest.

### Real use case

Walmart mines baskets for complementary product pairs and uses the relationships in recommendations.

### Advantage

Readable rules can support cross-selling without labeled outcomes.

### Limitation

Popularity and promotions distort co-occurrence. A relationship does not prove causation.

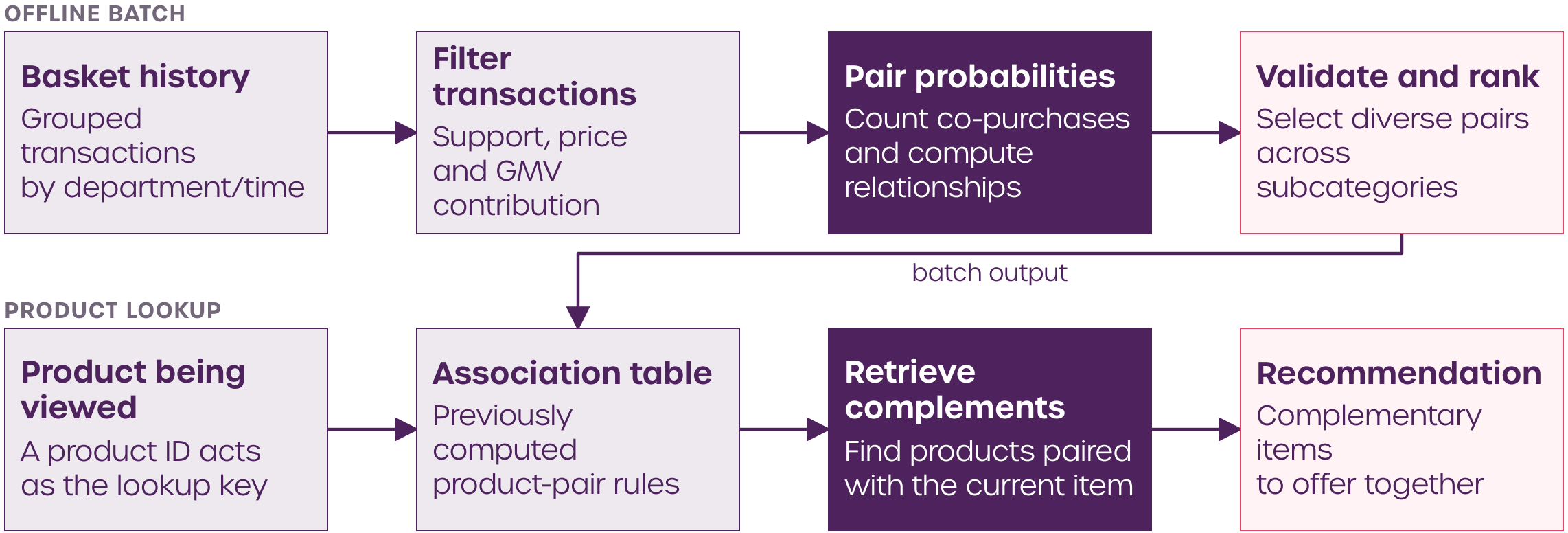
Walmart evaluated FP-Growth, then selected simpler pairwise co-occurrence rules for scale.

**Products bought together can inform products recommended together.**

Source: Walmart Global Tech

# Walmart: basket associations

Batch processing builds the product-pair lookup used by recommendations




The deployed winner uses pairwise probabilities. Walmart evaluated FP-Growth as an alternative.

Source: Walmart Global Tech

# Three classification outputs

| Task | Labels per observation | Illustrative output |
|---|---|---|
| Binary | One of two | Legitimate or bot traffic |
| Multiclass | One of several | One expense category |
| Multilabel | Several may apply | Several product interests |

"Multicategoría" is treated here as multilabel classification. Examples illustrate the output structure.

# Binary classification

## Cabify (Spain)

Operational service via Infobip Signals

### Concept

Assign one of two labels, often by applying a threshold to a risk score.

### Advantage

Automates high-volume decisions and reduces manual intervention.

### Real use case

Infobip Signals distinguishes legitimate SMS from bot traffic and blocks fraudulent requests for Cabify.

### Limitation

False positives can block real customers. Attack patterns change over time.

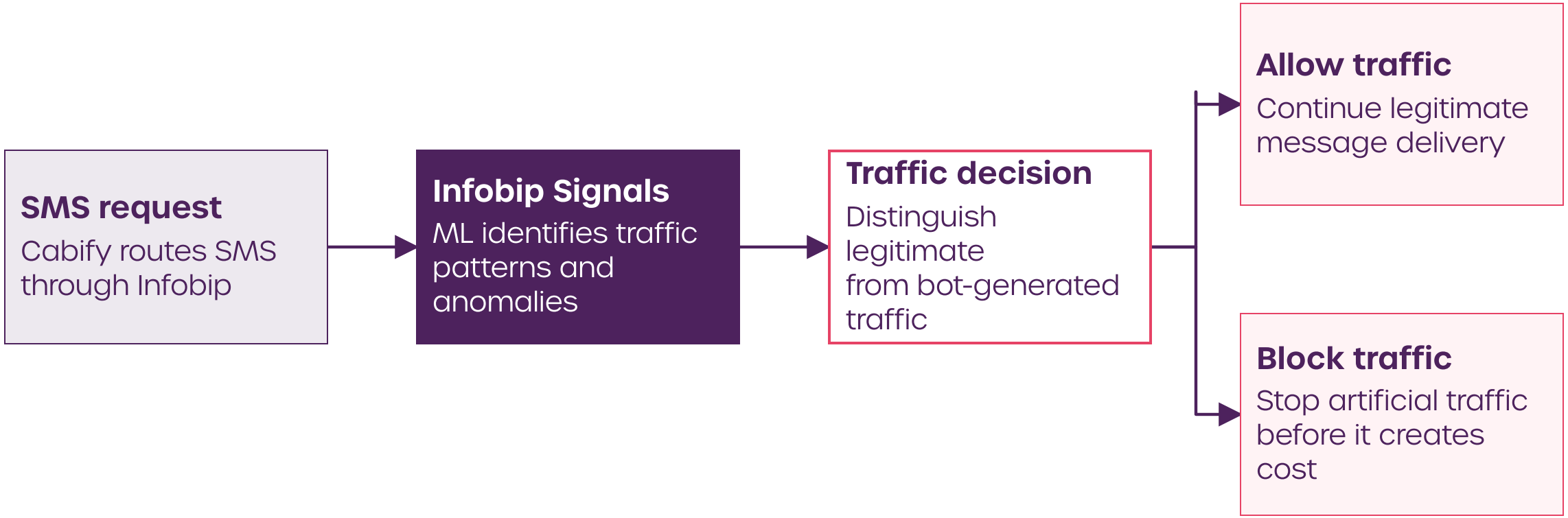
Infobip reports 15.7% less AIT-related financial loss in one quarter. Model architecture is undisclosed.

**A two-way prediction becomes useful when its errors have clear business costs.**

Source: Infobip customer case with Cabify testimony

# Cabify: legitimate or artificial traffic

The ML service makes a concrete allow/block decision on SMS traffic




Infobip discloses the decision and use of ML, but does not publish features, model family or thresholds.

Source: Infobip customer case with Cabify testimony

# Multiclass classification

## Uber (United States)

COTA support assistant in production (2018 account)

### Concept

Choose one category from three or more mutually exclusive alternatives.

### Advantage

Reduces the effort of navigating a large support taxonomy.

### Real use case

Uber COTA predicts support issue types from ticket text and trip context, helping agents select the right category.

### Limitation

Similar issues and rare categories can confuse the classifier.

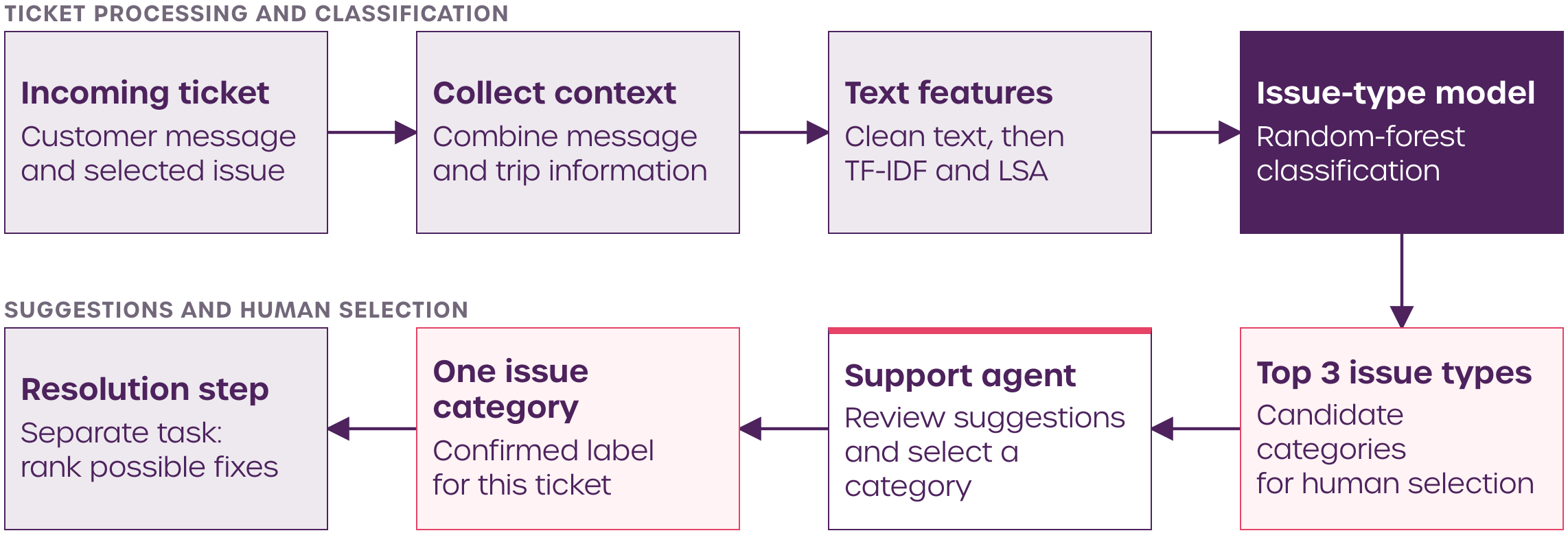
Issue identification is separate from COTA's solution-ranking task.

A ticket gets one issue category, with a human confirming the model's suggestions.

Source: Uber Engineering

# Uber COTA: identifying a support issue

Ticket text and trip context help a support agent select one issue category




Issue-type identification is the multiclass task. COTA ranks possible solutions in a separate step.

Source: Uber Engineering

# Multilabel classification

## Amazon Retail (United States)

Production customer propensity inference

### Concept

Assign zero, one or several labels to the same observation.

### Advantage

Represents overlapping interests without forcing a single segment.

### Real use case

Amazon predicts customer action propensities across thousands of product categories, allowing overlapping interests.

### Limitation

Rare labels and different thresholds make training and evaluation harder.

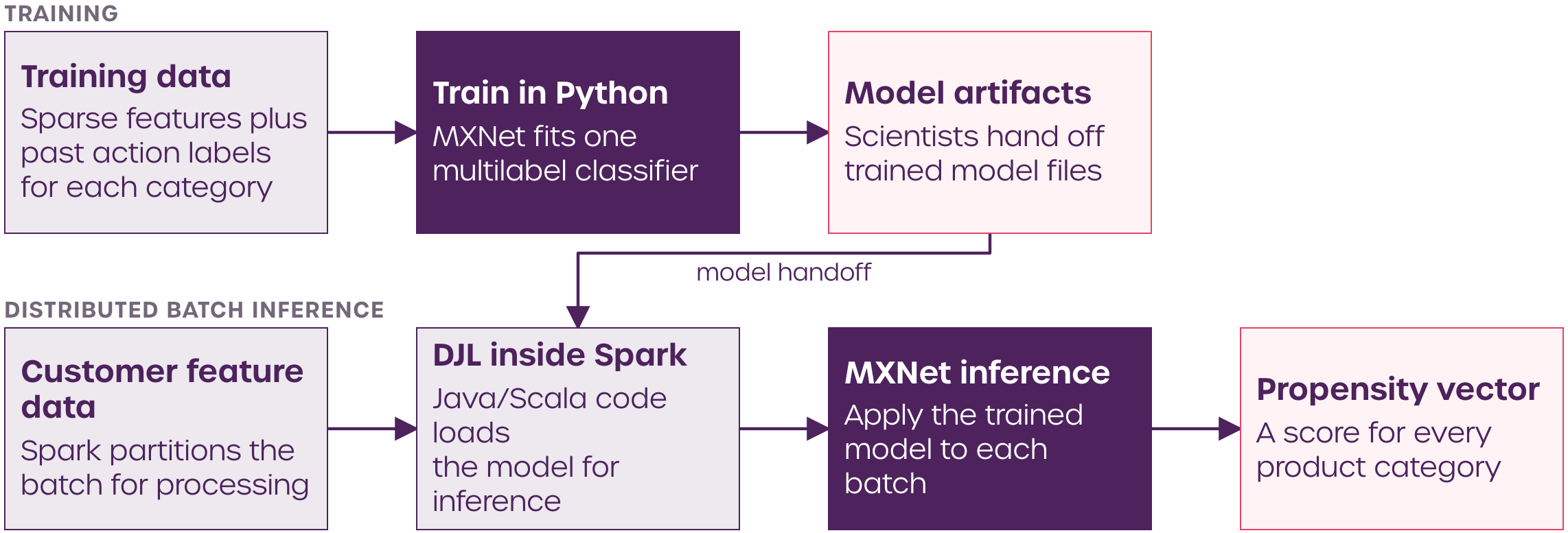
Amazon describes MXNet models with Spark and Deep Java Library inference.

**The same customer can belong to several useful categories at once.**

Source: Amazon / AWS Open Source Blog

# Amazon: thousands of propensities

One sparse customer vector feeds a shared model with many category outputs




Several categories can receive high propensity at the same time. The system does not select one winning label.

Source: Amazon / AWS Open Source Blog

# Clustering

## Spotify (Sweden)

Documented user-research method

### Concept

Discover groups with similar behavior without predefined group labels.

### Advantage

Finds unexpected segments and supports targeted user research.

### Real use case

Spotify combines UMAP with recursive HDBSCAN to discover finer segments of users.

### Limitation

Feature choices shape the groups. Stable clusters still need human validation.

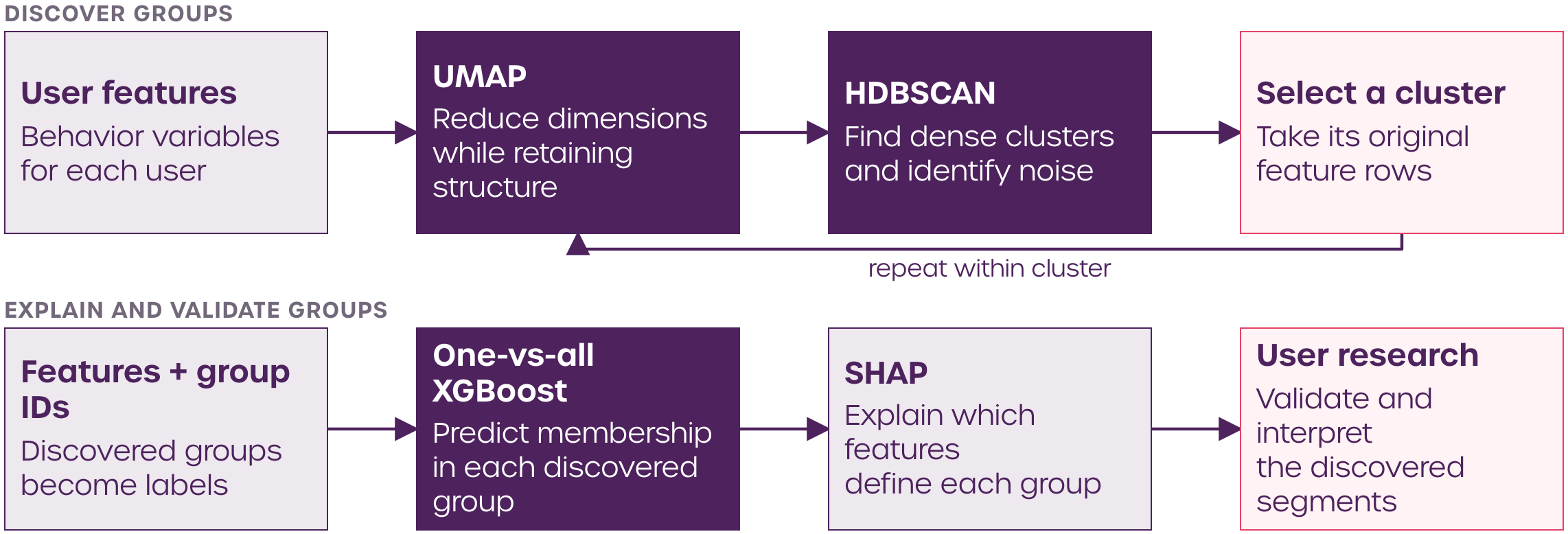
Spotify uses XGBoost and SHAP to explain the resulting segments.

**Clustering discovers groups that researchers can then investigate.**

Source: Spotify Engineering

# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster




Source: Spotify Engineering

# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

The people behind the use case

| | |
|---|---|
| **People** | Ana and Luis play Izal and Vetusta Morla. Marta plays Charlotte de Witte. |
| **Group** | Artist-play counts could group Ana and Luis together, apart from Marta. |
| **Why** | An explanation could identify Spanish indie listening as the group's main signal. |
| Possible use | Suggest Love of Lesbian to Ana if Luis listens to them. |

Illustrative artist example. The computed demo uses behaviour, not artist counts.

**Computed demo · synthetic listeners**

1 / 9

Technical details

## A concrete listening example

Ana and Luis share artists. Can we discover their group and explain what makes it different?

**Follow the same people**

Each next step explains what the algorithm would mean for Ana, Luis and Marta.

Illustrative story. The calculated charts use synthetic behaviour features, without artist counts or named users.

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering



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# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

## Original features

$X \in \mathbb{R}^{684 \times 6}$

Unlabelled observations

| Listener | Minutes | Sessions | Skips | Discovery | Playlists | Night |
|---|---|---|---|---|---|---|
| L001 | 222 | 11.4 | 10% | 17% | 52% | 72% |
| L002 | 124 | 6.6 | 46% | 64% | 41% | 29% |
| L003 | 32 | 1.8 | 28% | 27% | 75% | 8% |
| L004 | 127 | 6.6 | 42% | 66% | 32% | 36% |

Numeric features, no prescribed segment labels

684 rows × 6 features

StandardScaler $z_{ij} = (x_{ij} - \mu_j) / \sigma_j$ Fit once on the full demo population

Computed demo · synthetic listeners

2 / 9

Technical details

## An original feature vector for every listener

Ana's listening becomes a numeric row. Here we compute six behaviour features. Artist counts would be extra inputs.

**Listener L001**

| | | | |
|---|---|---|---|
| Minutes | 222 | Sessions | 11.4 |
| Skips | 10% | Discovery | 17% |
| Playlists | 52% | Night | 72% |

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering



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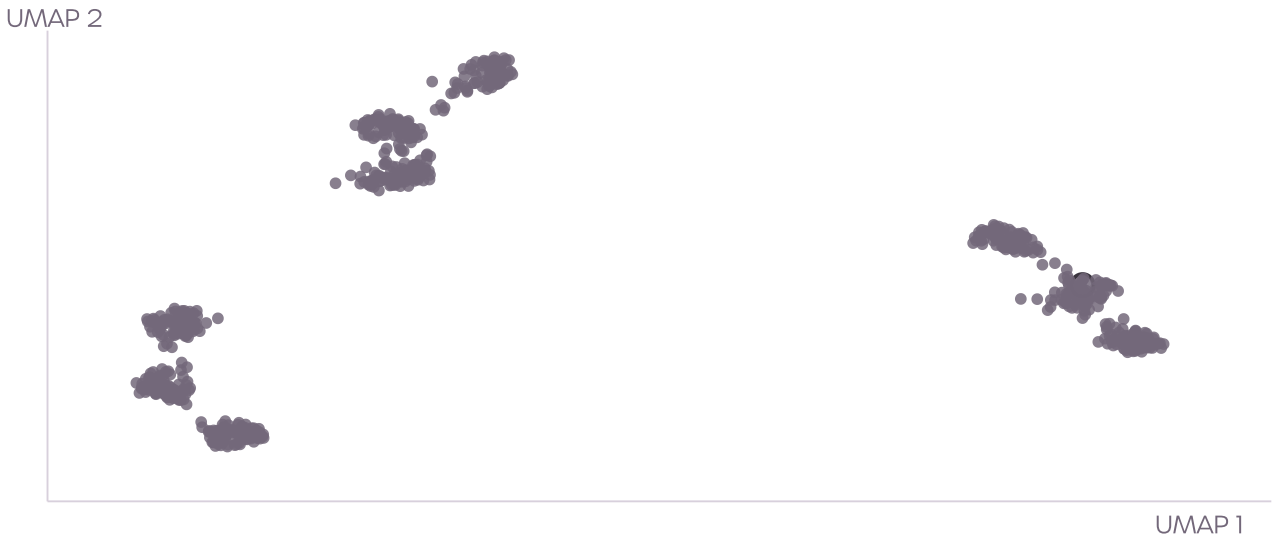
# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



## Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners




Display transition only. Point movement does not reproduce the optimizer's iterations.

3 / 9

Technical details

## UMAP makes neighbourhoods visible

With suitable artist features, Ana and Luis could appear nearby because they share artists. UMAP displays similarity, not a genre label.

$684 \times 6 \rightarrow 684 \times 2$

10-neighbour trustworthiness: 0.975

Local-neighbourhood diagnostic, not segment validity.

**Computed demo · synthetic listeners**

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

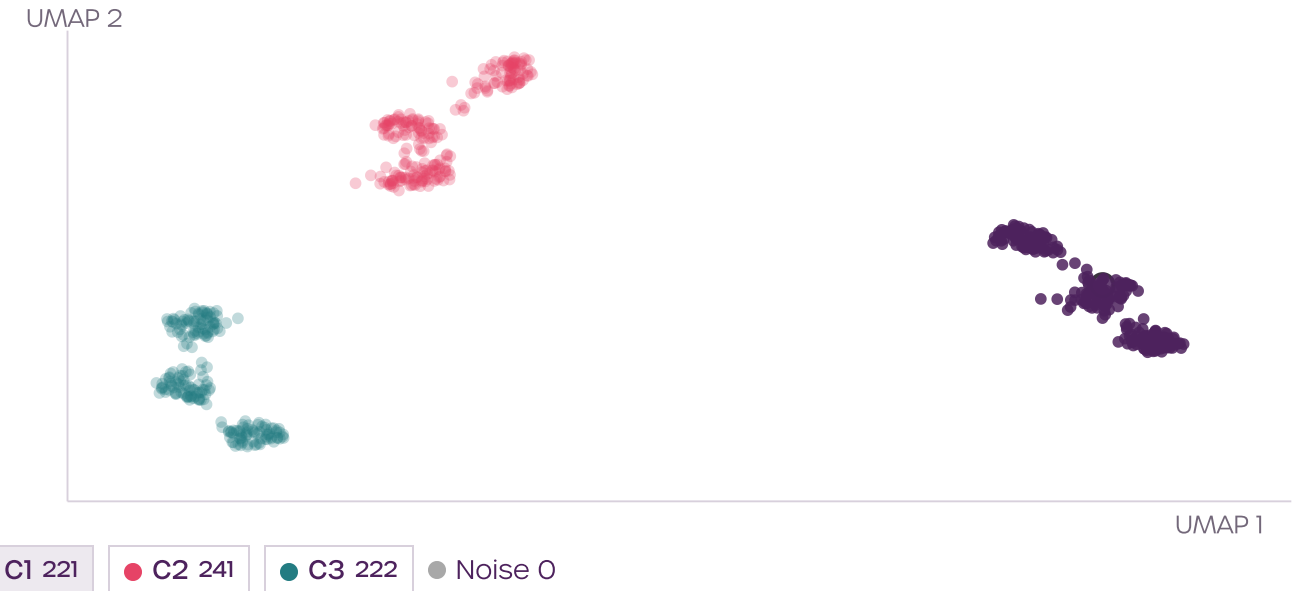
# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

## Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners




C1 221 | C2 241 | C3 222 | Noise 0

Computed demo · synthetic listeners

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Technical details

## HDBSCAN discovers density-based groups

A dense group could contain Ana and Luis, while Marta falls elsewhere. The colours show discovered groups. Our chart uses synthetic behaviour data.

Inspect group: C1 · 221 listeners

**32.3% of this node**

Noise: 0 / 684

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

# Spotify: discovering and explaining groups

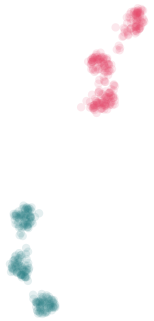
The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 **Re-embed** | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

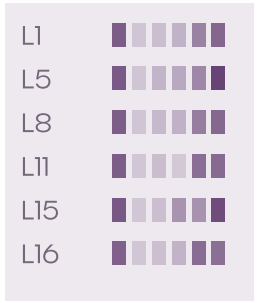
## A new fit, not a magnified plot

**1. Select C1**



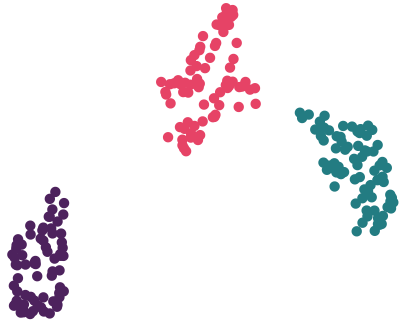


**2. Original X rows**




221 × 6 features

**3. Refit both models**



3 groups · 0 noise

New coordinates come from a new UMAP fit on the selected original rows.

**Computed demo · synthetic listeners**

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Technical details

## Recursion starts from the original feature rows

Analyse Ana's group again using its original rows. This could reveal finer habits within her broader group.

Parent cluster: C1 · 221 rows

Inspect C1 child fit

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering



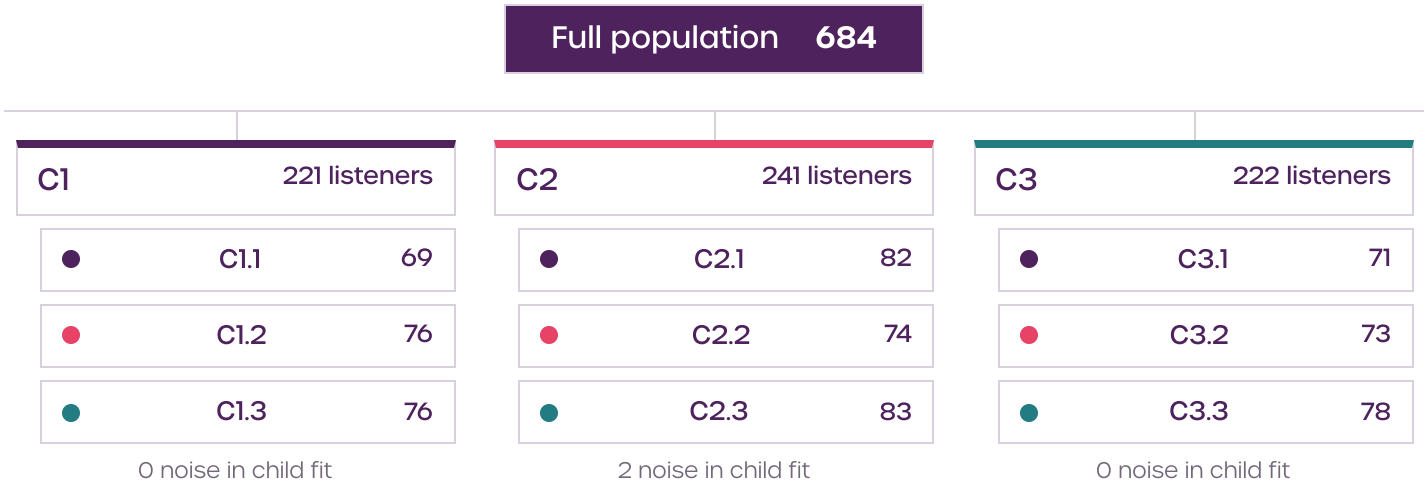
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# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



## Computed cluster hierarchy




Noise stays unassigned. Child counts can sum to less than their parent.

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Technical details

## The hierarchy records each discovery step

Ana's broad group could split into subgroups. The tree records which users enter each new fit. Click a branch to inspect its computed population.

**4 computed fits**

Each child has its own UMAP coordinates and HDBSCAN assignments.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

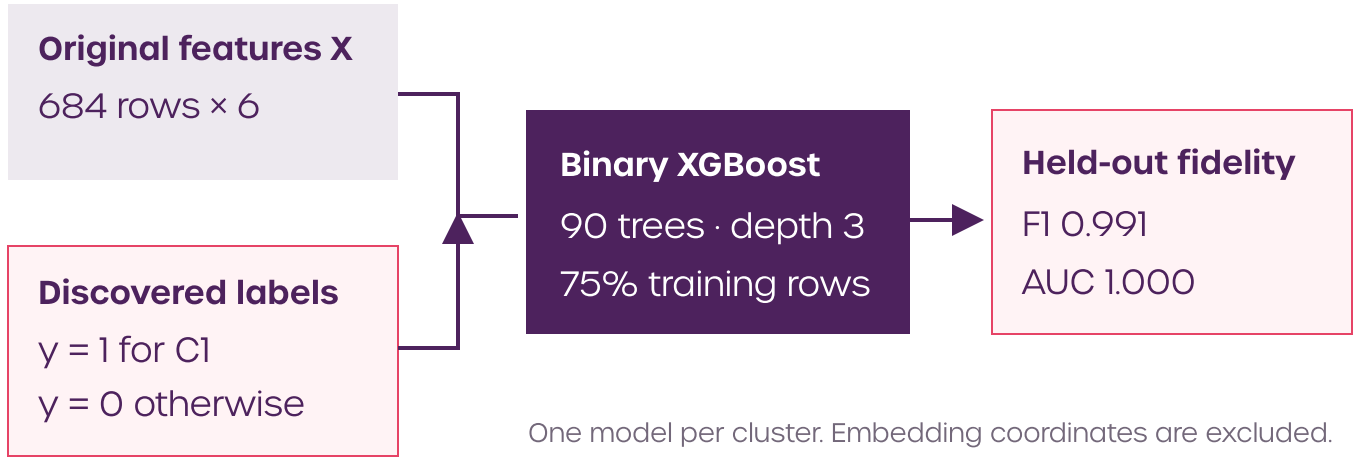
Source: Spotify Engineering

# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



## One-versus-rest training




One model per cluster. Embedding coordinates are excluded.

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Technical details

## A supervised model learns discovered membership

XGBoost learns to predict Ana's group versus everyone else, using the original features and discovered labels.

Membership model: C1 versus rest

513 training / 171 held-out

The split tests label reproduction, not new-user cluster discovery.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

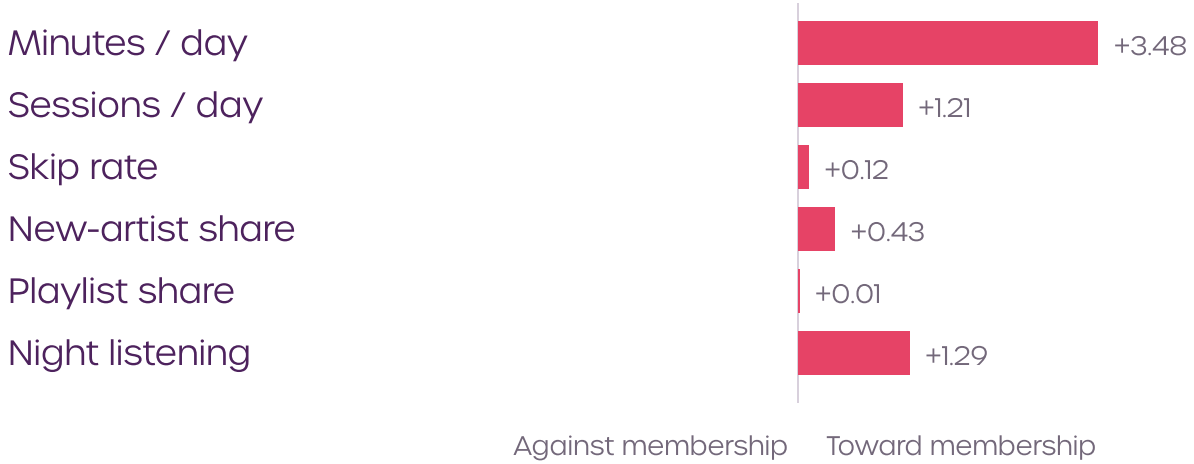
# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



## Calculated TreeSHAP contributions

This listener | Mean |SHAP| | Signed contribution to membership log-odds




Base -0.80 + Σ SHAP 6.55 = log-odds 5.75

**P(C1) = 99.7%**

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Technical details

## SHAP explains the membership predictor

SHAP explains a listener's group score. Indie listening could matter with artist inputs. These bars explain behaviour.

Explain membership in: C1

Listener: L001 · held-out

Held-out listener. Independent binary scores need not sum to 1.

**Computed demo · synthetic listeners**

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

# Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

Why interpretation is needed

| | |
|---|---|
| **People** | Ana and Luis play Izal and Vetusta Morla. Marta plays Charlotte de Witte. |
| **Group** | Artist-play counts could group Ana and Luis together, apart from Marta. |
| **Why** | An explanation could identify Spanish indie listening as the group's main signal. |
| Possible use | Suggest Love of Lesbian to Ana if Luis listens to them. |

Illustrative artist example. The computed demo uses behaviour, not artist counts.

Computed demo · synthetic listeners

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Technical details

## What the results mean

Group C1 Listener L001

### Across listeners

Minutes / day and Night listening have the largest average effects.

### This listener

Minutes / day = 222 is the strongest push toward C1.

Model explanations do not prove causes or validate segments.

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering



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# Anomaly detection

## Adyen (Netherlands)

Production payment-volume monitoring

### Concept

Flag observations that differ from the expected pattern for their context.

### Advantage

Can identify rare operational problems without labeled incident examples.

### Real use case

Adyen models merchant payment volumes and alerts teams when observations fall outside expected bounds.

### Limitation

Holidays and merchant changes cause false alerts. Sensitive thresholds increase noise.

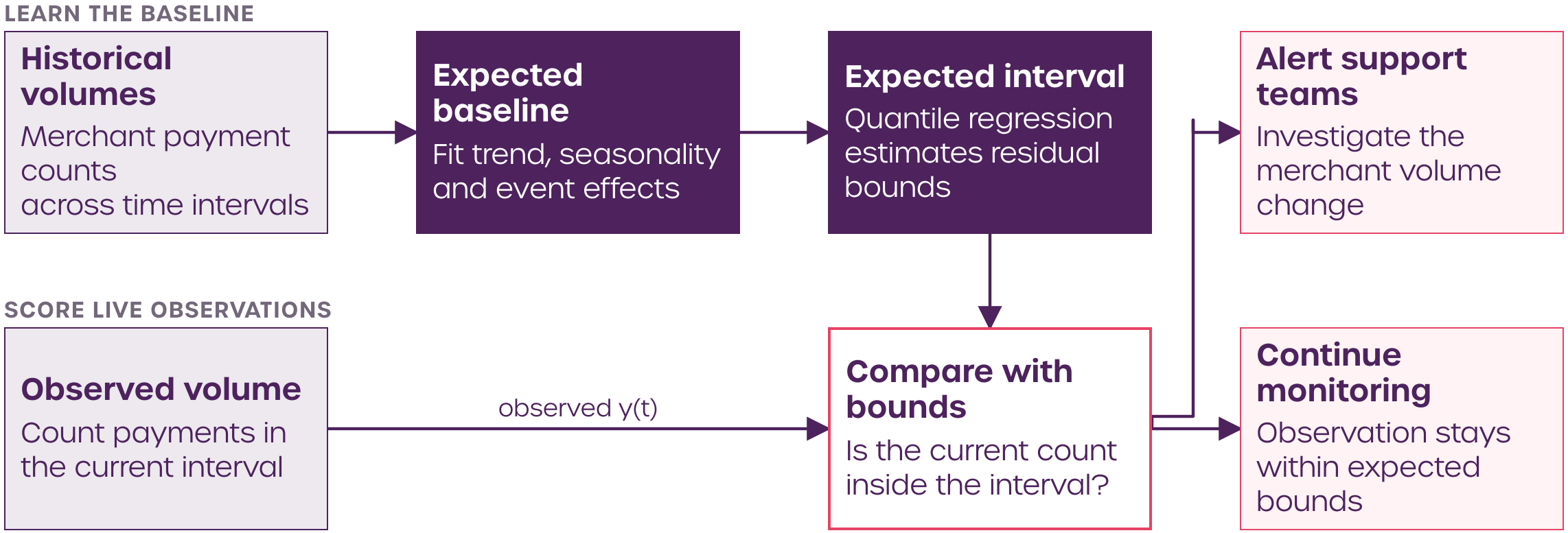
The model captures trend, seasonality and events, with quantile regression for uncertainty bounds.

**An anomaly is unusual relative to a meaningful baseline.**

Source: Adyen Engineering

# Adyen: detecting a payment-volume drop

A contextual baseline and uncertainty bounds define what counts as unusual




A low absolute count can be normal. The decision depends on the merchant, time and expected interval.

Source: Adyen Engineering

# Regression analysis

## idealista/data (Spain)

Commercial automated valuation service

### Concept

Estimate a continuous number from explanatory features.

### Real use case

idealista/data combines comparables and hedonic ML to estimate property values from location and attributes.

### Advantage

Produces consistent estimates across large property portfolios.

### Limitation

Missing attributes and market changes reduce accuracy. Estimates differ from completed sale prices.

The company discloses the valuation approach, but not the exact estimator.

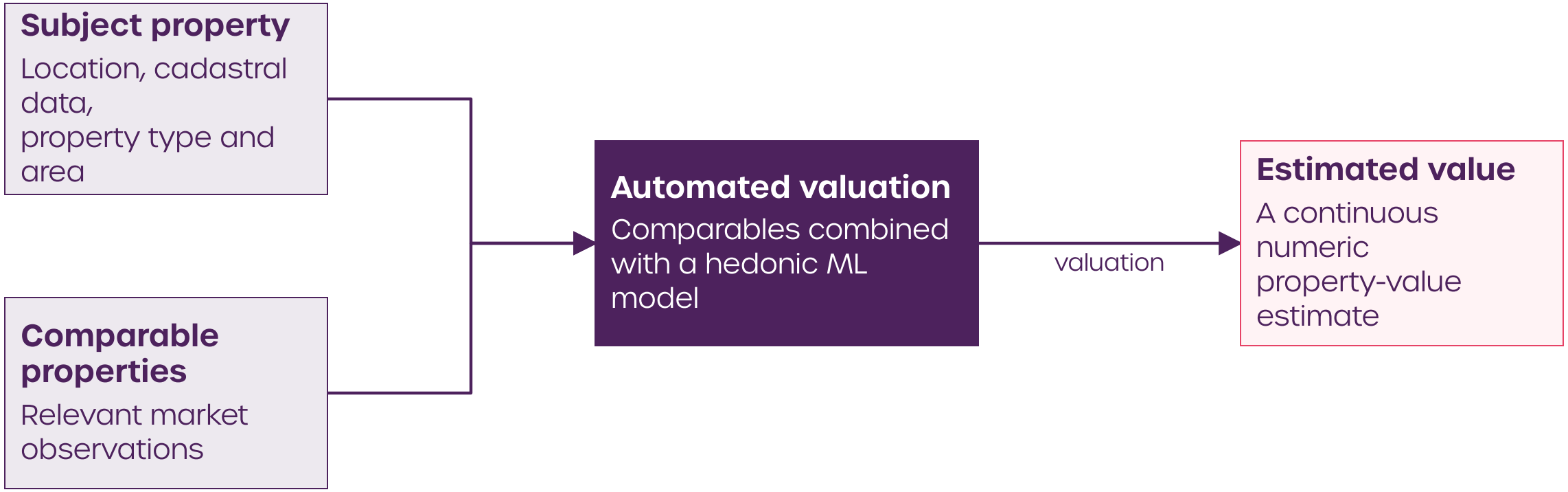
**Regression estimates a number from the characteristics of an observation.**

Source: idealista/data and idealista labs

# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data




Source: idealista/data and idealista labs



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# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

The subject property becomes a feature vector

## Ana's apartment

Zone B · fictional market

| | | | |
|---|---|---|---|
| Floor area | 80 m² | Bedrooms | 2 |
| Renovated | Yes | Lift | Yes |

↓

$$x = [80, 2, 1, 0, 1, 1]$$

area, beds, zone B, zone C, renovated, lift

Zone A is the reference. Zones B and C use separate 0/1 indicators.

Computed demo · synthetic homes and prices

1 / 5

Technical details

### Ana's home to value

Ana wants to estimate her apartment's price. Change its attributes to follow the same home through every step.

Floor area — 80 m²

Bedrooms — 2

Fictional zone — Zone B

Renovated — ☑

Lift — ☑

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs



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# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

## Comparable evidence from the training set

Zone B only · 80 m² · 2 bedrooms · six closest matches

| Property | Area | Beds | Renovated | Lift | Reference price |
|---|---|---|---|---|---|
| P425 | 85 m² | 2 | Yes | Yes | €302,180 |
| P276 | 91 m² | 2 | Yes | Yes | €339,450 |
| P201 | 80 m² | 3 | Yes | Yes | €311,444 |
| P393 | 80 m² | 1 | Yes | Yes | €283,257 |
| P275 | 85 m² | 3 | Yes | Yes | €311,032 |
| P216 | 71 m² | 2 | No | Yes | €278,683 |

Mean of these six prices **€304,341**

All properties and prices are synthetic. Matching rules are for teaching.

**Computed demo · synthetic homes and prices**

2 / 5

Technical details

## Six similar reference properties

Find six homes like Ana's: the same zone, similar size and bedrooms. Their reference prices provide a first comparison.

### What makes a match?

Same zone, similar area and bedrooms, plus renovation and lift.

Only training properties enter this comparison.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

Learning first, prediction second

**TRAINING · computed once**

**360 known examples**
Features X + price y

→ **Linear regression**
Learn weights β

→ **OLS fit**
$\min \Sigma(y - \hat{y})^2$

**PREDICTION · updates with your inputs**

**This home's features**
80 m² · Zone B

→ **Fixed fitted model**
$\hat{y} = \beta_0 + \Sigma \beta_j x_j$

→ **Model price**
**€305,790**

Computed demo · synthetic homes and prices

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Technical details

## Regression learns a numeric relationship

Learn from known homes, then predict Ana's price from her apartment's attributes.

### What does it learn?

For example, +10 m² adds €19,802 to this model's prediction, holding other inputs fixed.

This is a fitted association in synthetic data.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

## A transparent blend for this demo

Similar-home mean

**€304,341**

50% of the demo estimate

+

Regression estimate

**€305,790**

50% of the demo estimate

Illustrative estimated price

**€305,000**

Rounded to the nearest €1,000

The 50/50 blend is our teaching choice. idealista's production weighting is not public.

Computed demo · synthetic homes and prices

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Technical details

## One continuous price estimate

For Ana's home, combine the similar-home mean with the regression prediction. This demo uses an illustrative 50/50 average.

**Held-out mean absolute error**

€10,577 across 120 synthetic homes.

A demo evaluation, not idealista's accuracy or an individual prediction interval.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

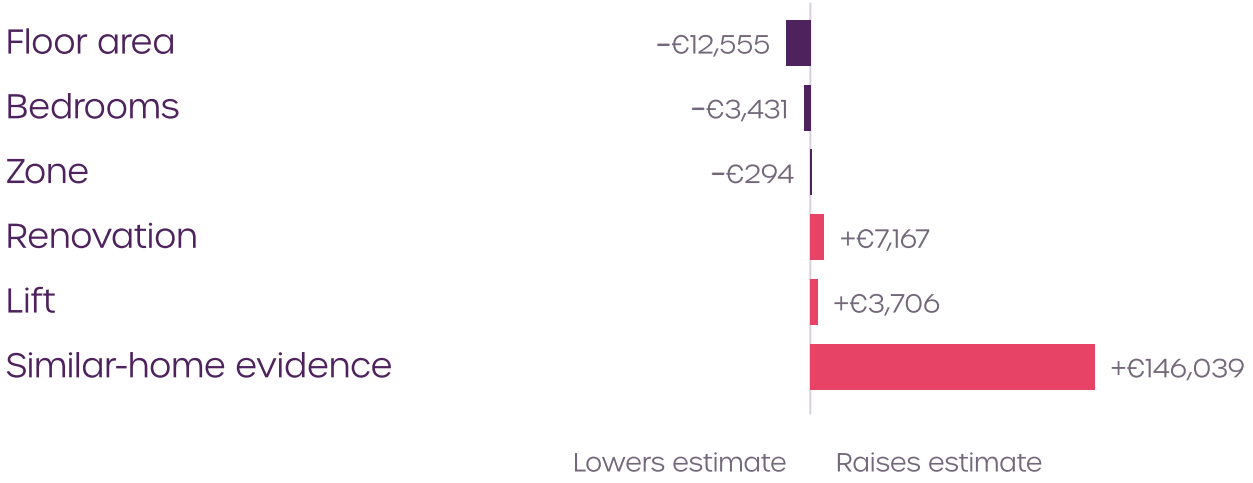
# idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

## An arithmetic breakdown of this estimate




Baseline €316,604 + adjustments = €305,065

Exact demo arithmetic, before rounding. The baseline is the mean training-model prediction.

**Computed demo · synthetic homes and prices**

5 / 5

Technical details

## What problem does this solve?

For Ana, explain how size, renovation and lift affect this estimate.

### This apartment

80 m² versus a training average of 93 m²: floor area contributes −€12,555.

### Renovation and lift

Renovated: +€7,167. Has a lift: +€3,706.

Bars explain this calculation. They do not prove causes or guarantee a sale price.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs



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# Forecasting

## Iberdrola (Spain)

Operational MeteoFlow system

### Concept

Estimate future values using time-dependent patterns and available drivers.

### Real use case

MeteoFlow combines weather and generation data, using ML to refine renewable output forecasts.

### Advantage

Supports planning when wind and solar output vary with weather.

### Limitation

Forecast uncertainty grows with the horizon and unexpected weather changes.

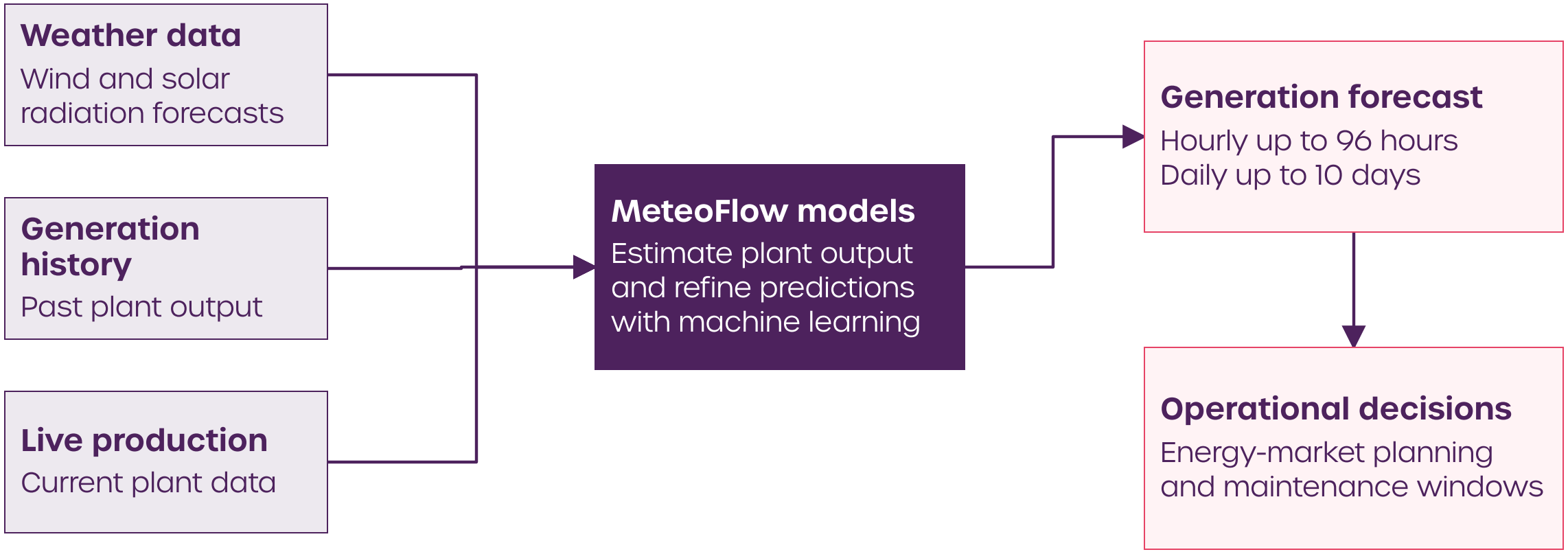
Iberdrola documents 450+ wind and solar plants, with hourly forecasts up to 96 hours.

**Forecasting turns an uncertain future into a quantified planning input.**

Source: Iberdrola

# Iberdrola: renewable output forecasts

MeteoFlow combines external weather with historical and live plant information




The data sources and forecast horizons are documented. Iberdrola does not name the exact ML model family.

Source: Iberdrola

# Sentiment analysis and NLP

## Booking.com (Netherlands)

Documented review-analysis workflow

### Concept

Extract opinions from language and link each opinion to what it describes.

### Advantage

Makes specific praise and complaints visible across many reviews.

### Real use case

Booking.com links review opinions to 50+ aspects, such as cleanliness or location.

### Limitation

Sarcasm, mixed opinions and language differences can change interpretation.

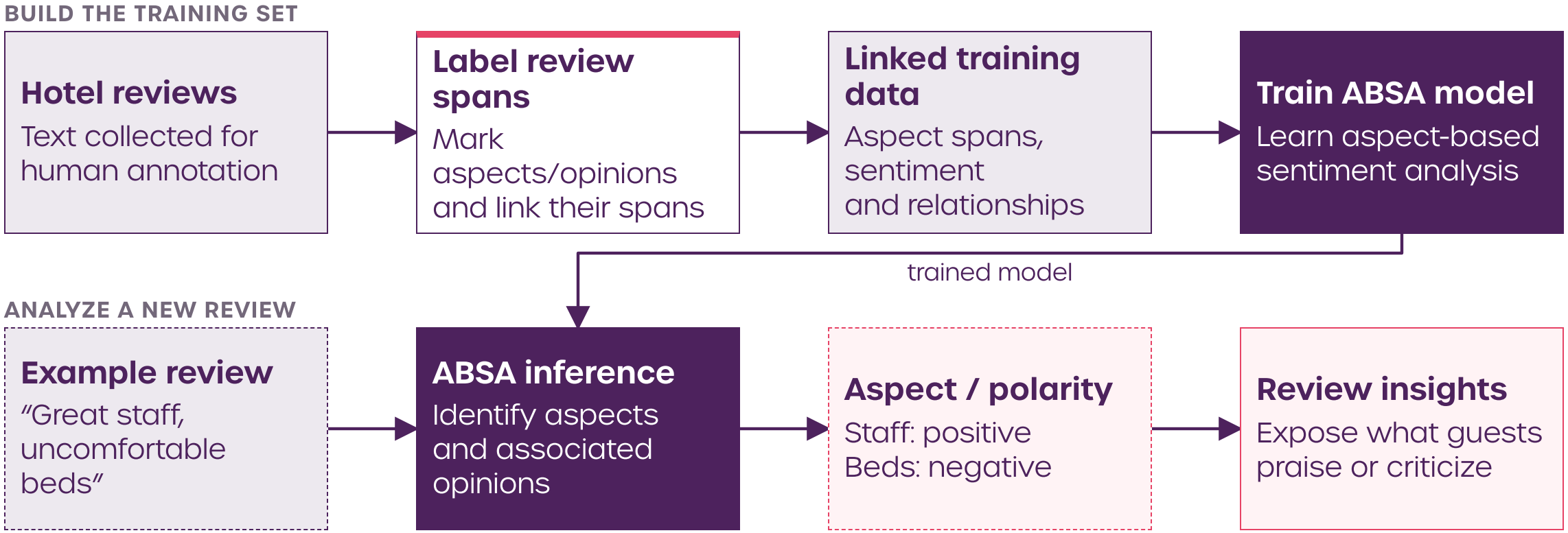
Aspect extraction identifies the subject. Sentiment analysis estimates the opinion about it.

**A review can praise the location while criticizing the room.**

Source: Booking.com team / AWS Machine Learning Blog

# Booking.com: aspect-level opinions

Training links aspect spans to sentiment. Inference returns separate judgments




Dashed boxes contain a teaching example. The article documents annotation and model purpose, not the model architecture.

Source: Booking.com team / AWS Machine Learning Blog

# Text mining

## ING (Netherlands)

Deployed credit-risk monitoring application

### Concept

Extract structured facts and recurring signals from a large text collection.

### Real use case

ING's Early Warning System combines signals from news and market data for credit-risk analysts.

### Advantage

Helps analysts monitor more external information than they can read manually.

### Limitation

Irrelevant or duplicated news creates noise. A signal still needs an analyst's judgment.

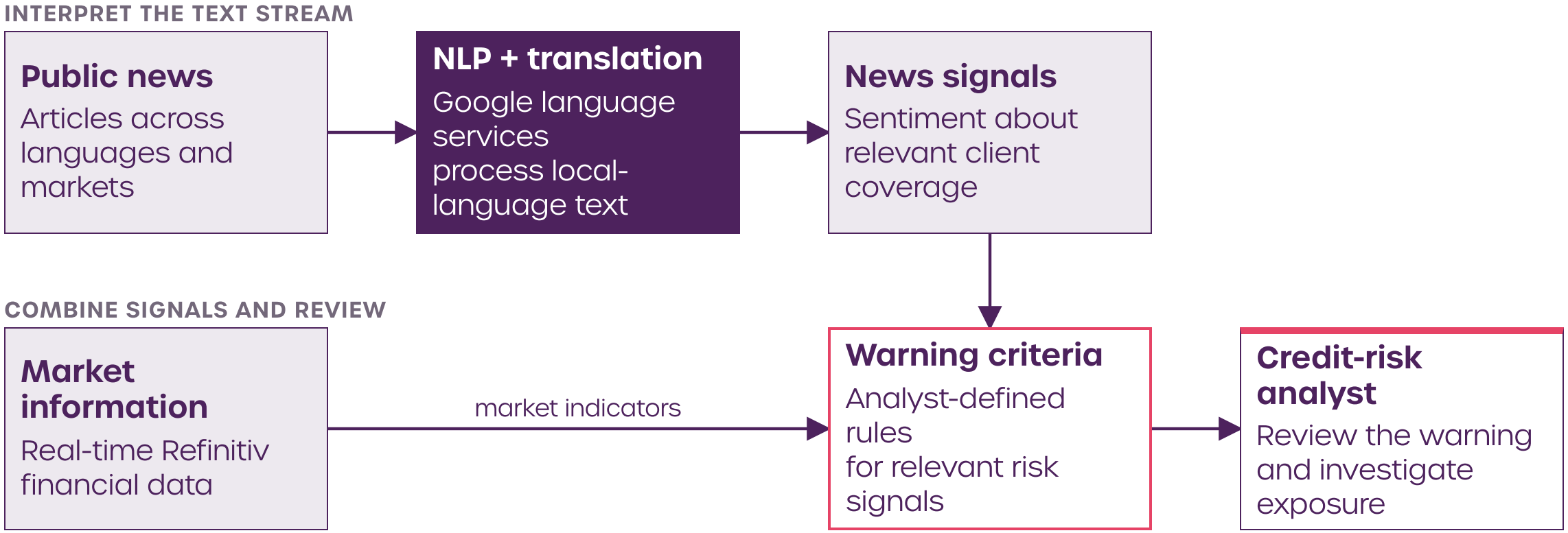
ING describes processing up to 80,000 news articles per day.

**Text mining turns a document stream into evidence for investigation.**

Source: ING

# ING: news becomes a risk watchlist

News analysis and market signals meet analyst-defined warning criteria




The output is an analyst warning. The source presents default prediction as a future ambition.

Source: ING

02

# Deep learning

Neural networks learn visual features and generate new images



Innoit

# Pattern recognition

## BMW Group (Germany)

Series-production applications since 2018

### Concept

Learn visual features that identify an object or a recurring configuration.

### Advantage

Performs repeated visual checks at production speed.

### Real use case

At Dingolfing, BMW compares a vehicle's model badge with its order specification and flags missing or incorrect badges.

Employees label component images and train a neural network.

### Limitation

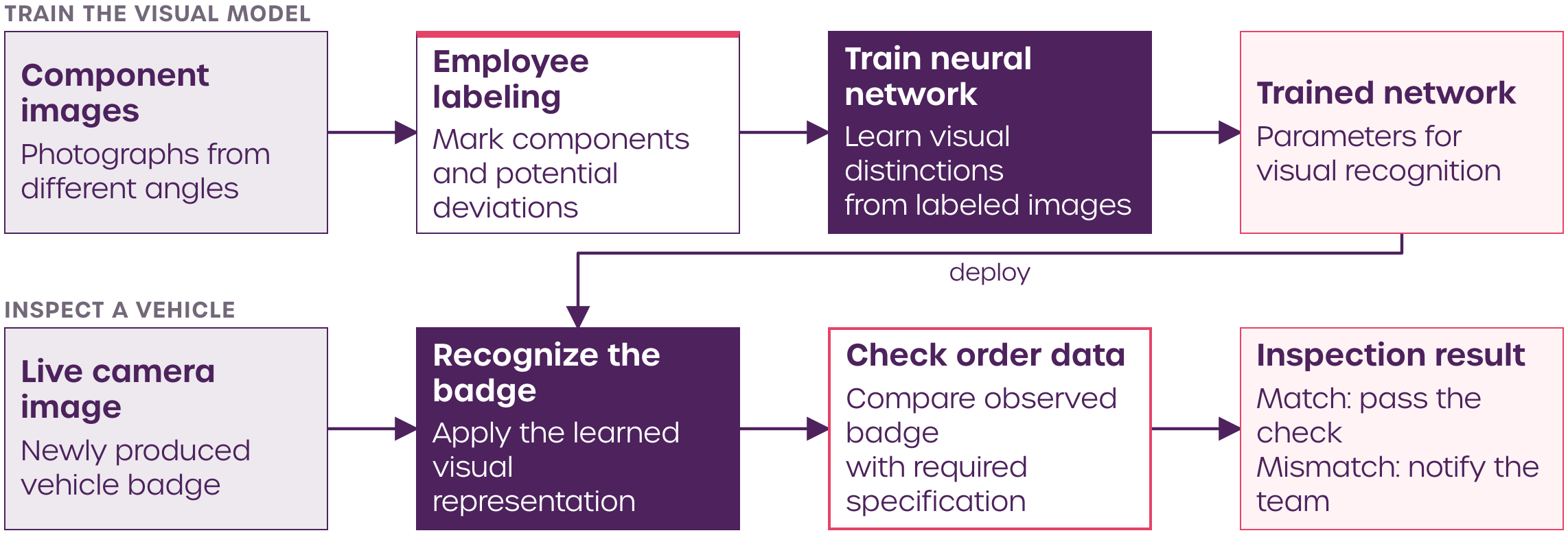
New components and lighting conditions require representative images and monitoring.

A learned visual pattern can turn a camera image into a quality check.

Source: BMW Group

# BMW: image recognition checks the badge

Learning the visual pattern and checking the order are separate steps




The neural network recognizes the image. Business logic compares the recognition result with vehicle order data.

Source: BMW Group

# BMW visual inspection

The camera observes a vehicle component.

**The neural network recognizes it**

Software compares the result with the order specification and flags a mismatch.



BMW Group production photo. Source: BMW Group, 15 July 2019.

# Image generation

## Mango (Spain)

Published Mango Teen campaign

### Concept

Generate new images using learned visual structure.

### Advantage

Expands creative options and supports repeatable production workflows.

### Real use case

Mango created Sunset Dream using real garment photos, a generative model and human selection and retouching.

### Limitation

Product fidelity and editorial quality still require human art direction.

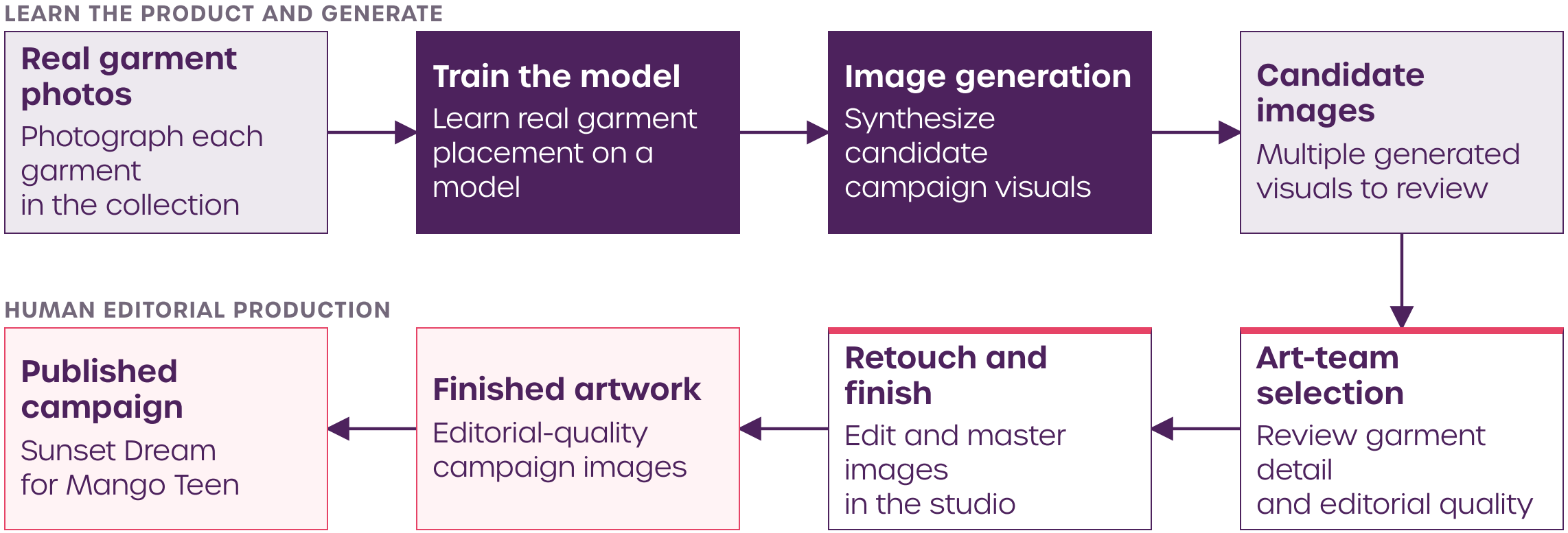
Mango names no specific model architecture. Diffusion and GANs are general examples, not confirmed choices.

**Generate new imagery while preserving the real product's characteristics.**

Source: Mango Fashion Group

# Mango: real garments to campaign images

The published workflow includes model training and human editorial production




Mango does not name diffusion, GANs or a specific model. Human selection and retouching are documented steps.

Source: Mango Fashion Group

# Mango Teen: Sunset Dream

A published campaign created with generative AI in July 2024.

**Real garment photographs anchor the work**

Mango's art team selects and retouches generated images to preserve product detail and editorial quality.



Official campaign image. Credit: Mango Fashion Group.

# 03

# GenAI systems

Retrieval, multimodal generation and coordinated tool use



Innoit

# What RAG adds to a language model

## Retrieval

1. Read the employee's question.
2. Search the authorized knowledge base.
3. Select relevant document passages.

## Grounded generation

4. Give the model the retrieved evidence.
5. Generate an answer with sources.
6. Check the answer or escalate.

# Intelligent search with RAG

## Allianz (Germany)

EKA live at Allianz Leben from July 2024

### Concept

Retrieve relevant knowledge before an LLM generates an answer.

### Advantage

Makes internal knowledge easier to access and inspect.

### Real use case

Allianz EKA searches AWISO, summarizes relevant knowledge and links call-center staff to full sources.

### Limitation

Missing retrieval results and outdated documents can still produce wrong answers.

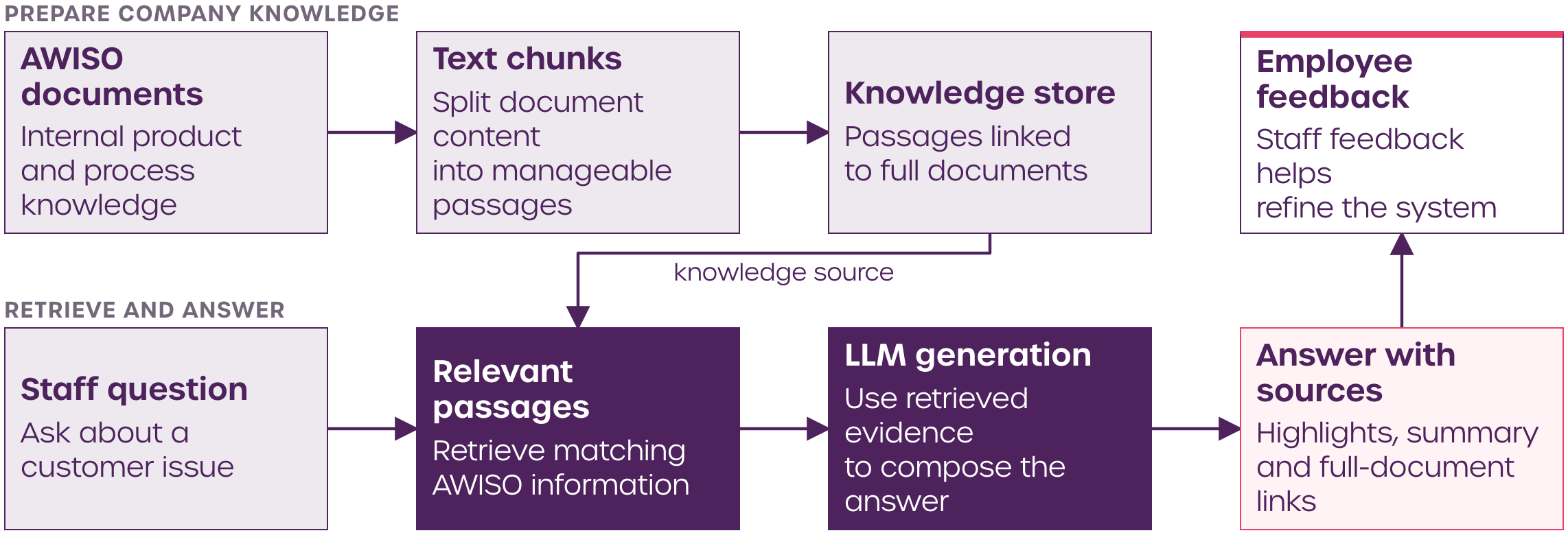
Allianz reports over 90% accuracy on common queries, without publishing the benchmark methodology.

**Retrieve the evidence first, then generate a grounded answer.**

Source: Allianz

# Allianz: retrieval grounds the answer

EKA searches AWISO knowledge and returns inspectable information to staff




This functional RAG view omits EKA's other agents. The source does not publish its exact retriever or embedding model.

Source: Allianz

# Multimodal generation

## Google (United States)

Veo 3 launched in Google Vids (June 2025)

### Concept

Generate related content across modalities, such as video and sound.

### Real use case

Google Vids uses Veo 3 to turn a text prompt into a video clip with sound, ready to preview and insert.

### Advantage

Creates an audiovisual draft without filming or recording a first version.

### Limitation

Visual continuity, speech and factual details still need human review.

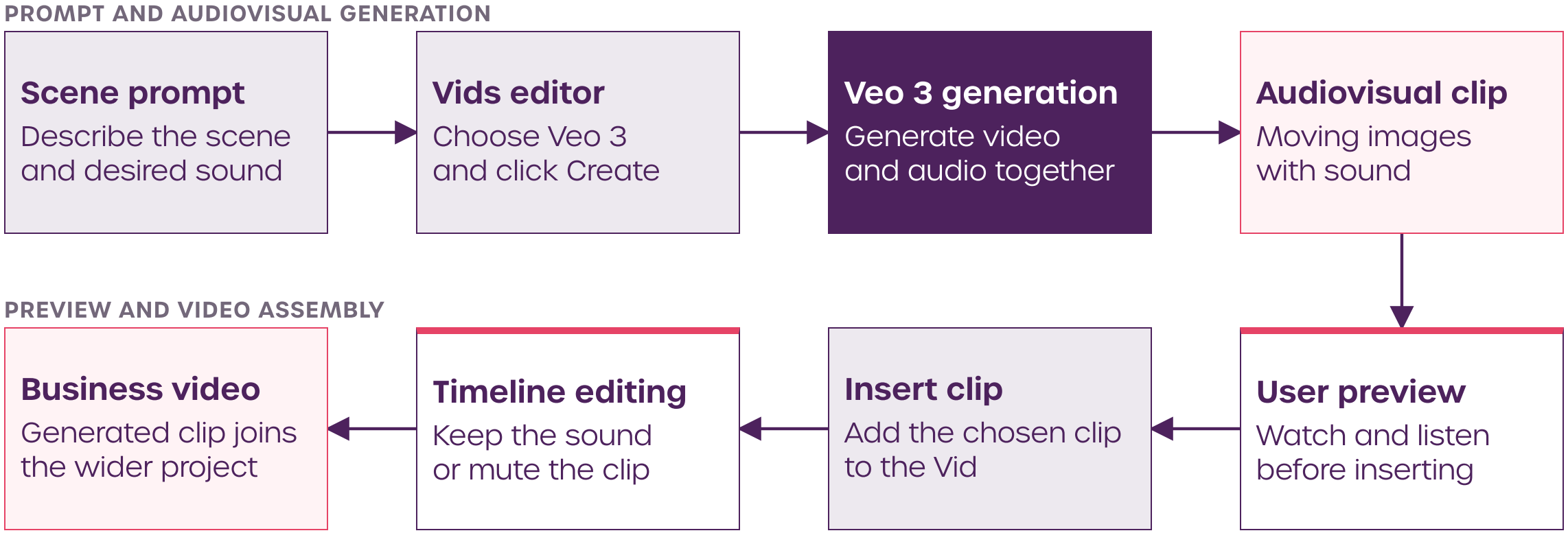
The launch release documents 8-second clips. This is a shipped product capability.

**One prompt can generate both the moving scene and its accompanying sound.**

Source: Google Workspace Updates

# Google Vids: generating video and sound

Veo 3 generates both modalities. The user previews the clip before inserting it




The June 2025 release documents 8-second clips with sound. Preview and insertion remain user-controlled.

Source: Google Workspace Updates

# What an orchestrator coordinates

## Request and routing

1. Interpret the goal and context.
2. Select a relevant specialist or tool.
3. Request clarification when needed.

## Action and feedback

4. Execute the authorized operation.
5. Inspect the result and update state.
6. Reply, retry or hand over to a person.

# Agentic workflow orchestration

## BBVA (Spain)

Blue assistant deployed in Spain in 2025

### Concept

Coordinate specialized agents and tools to complete a bounded task.

### Advantage

Connects conversational requests to defined account and card operations.

### Real use case

Blue routes requests to a banking tool, an information agent or a clarification step.

### Limitation

Routing errors can select the wrong workflow. Permissions and confirmations matter.

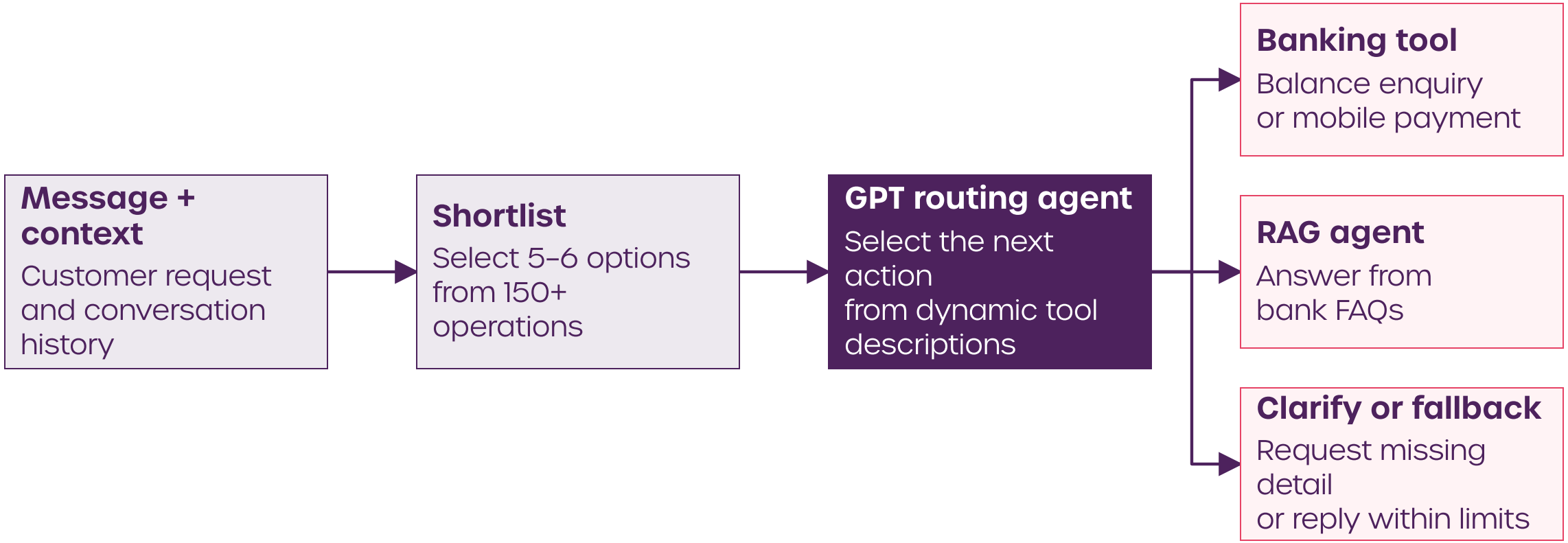
BBVA describes filtering 150+ operations to around 5–6 candidates before routing.

**Agents become operationally useful when the right tool can carry out the task.**

Source: BBVA AI Factory and BBVA rollout announcement

# BBVA Blue: routing a banking request

The routing agent selects a specialist, a tool or a clarification response




Banking operations are callable tools. A conversation can branch between information, action and clarification.

Source: BBVA AI Factory and BBVA rollout announcement

# Method selection: structured outputs

| Business question | Method | Case |
|---|---|---|
| What occurs together? | Association rules | Walmart |
| Which of two outcomes? | Binary classification | Cabify |
| Which single category? | Multiclass classification | Uber |
| Which overlapping labels? | Multilabel classification | Amazon |
| Which groups emerge? | Clustering | Spotify |
| What deviates from normal? | Anomaly detection | Adyen |
| What numeric value? | Regression | idealista |
| What future value? | Forecasting | Iberdrola |

The output and business decision determine the task.

# Method selection: language and GenAI

| Business question | Method | Case |
|---|---|---|
| What is the opinion about? | Sentiment and NLP | Booking.com |
| What signals occur in text? | Text mining | ING |
| What visual pattern is present? | Pattern recognition | BMW |
| What image can we create? | Image generation | Mango |
| What do our documents say? | Search with RAG | Allianz |
| Can a prompt yield video and sound? | Multimodal generation | Google |
| Can software complete the task? | Agent orchestration | BBVA |

Image generation uses deep learning and is already a form of GenAI.

# Business value and model quality

A useful model changes a decision or makes a workflow easier.

- **Classification:** precision and recall reflect different error costs.
- **Prediction:** backtests need realistic time and data boundaries.
- **GenAI systems:** correct retrieval or fluent text alone does not prove task success.
- **Every method:** deployment needs ownership, monitoring and a clear fallback.

# Sources 1 of 3

**Walmart** Association rules

Walmart Global Tech · 24 Aug 2022

**Cabify** Binary classification

Infobip customer case with Cabify testimony · Undated customer case

**Uber** Multiclass classification

Uber Engineering · 3 Jan 2018

**Amazon Retail** Multilabel classification

Amazon / AWS Open Source Blog · 14 Jul 2020

**Spotify** Clustering

Spotify Engineering · 5 Dec 2023

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

# Sources 2 of 3

**Adyen** Anomaly detection

Adyen Engineering · 9 Oct 2019

**idealista/data** Regression analysis

idealista/data and idealista labs · Product page and 17 Feb 2021 engineering post

**Iberdrola** Forecasting

Iberdrola · Undated corporate project page

**Booking.com** Sentiment analysis and NLP

Booking.com team / AWS Machine Learning Blog · 14 Jan 2022

**ING** Text mining

ING · 17 Dec 2018

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

# Sources 3 of 3

**BMW Group** Pattern recognition

BMW Group · 15 Jul 2019

**Mango** Image generation

Mango Fashion Group · 10 Jul 2024

**Allianz** Intelligent search with RAG

Allianz · 7 Feb 2025

**Google** Multimodal generation

Google Workspace Updates · 9 Jun 2025

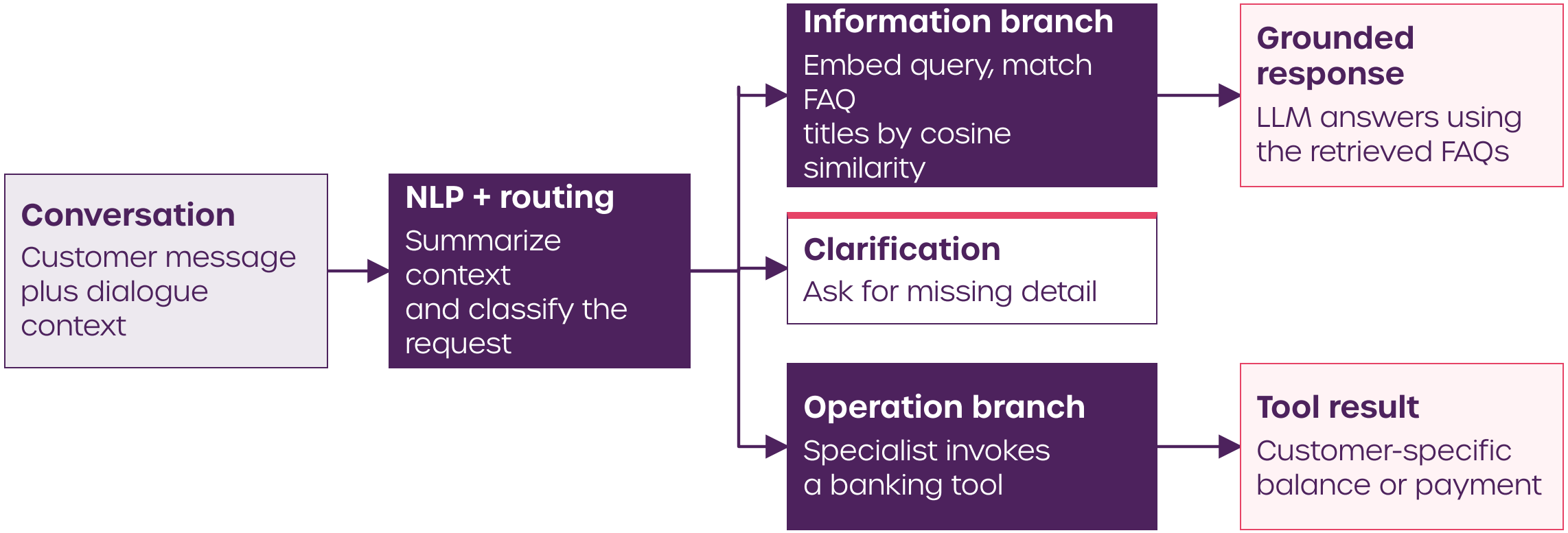
**BBVA** Agentic workflow orchestration

BBVA AI Factory and BBVA rollout announcement · 13 Nov 2025 technical account

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

# BBVA Blue: knowledge and actions

One assistant combines NLP, intent classification, RAG and tool orchestration




Generic FAQs and customer-specific operations follow different branches. Missing information triggers clarification.

Source: BBVA AI Factory: routing and RAG information agent

# Discussion

**Which business decision needs a better input?**

A rule, a prediction, generated content or a completed task?

www.inno-it.es